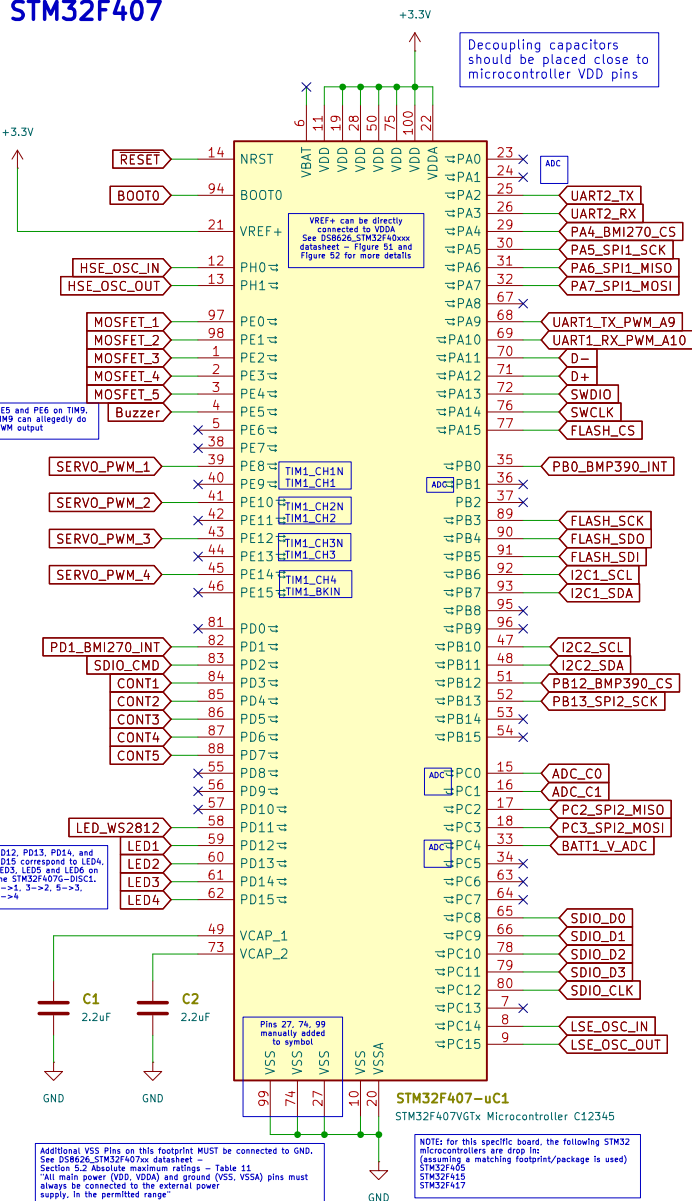
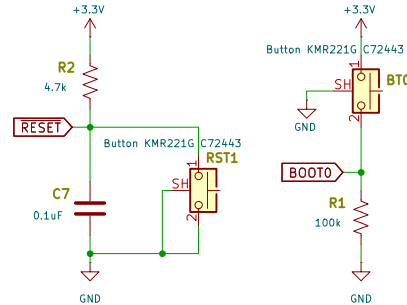


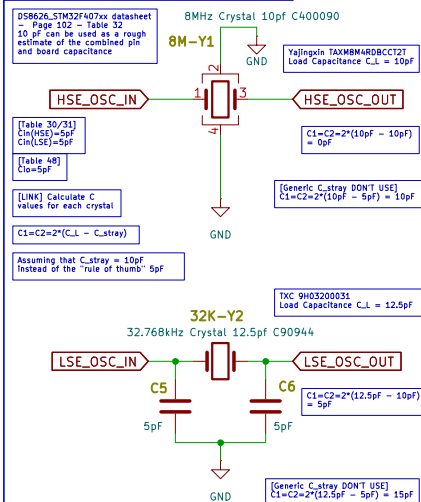
STM32F407



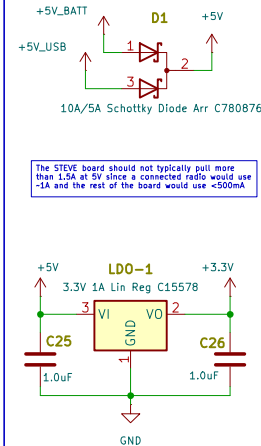
Reset and Boot



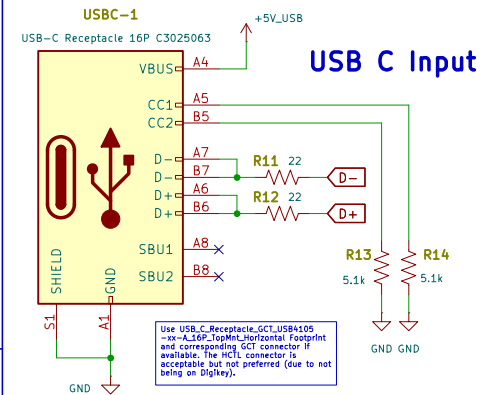
Crystals



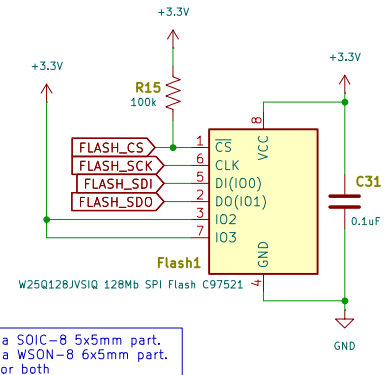
5V & 3.3V Out



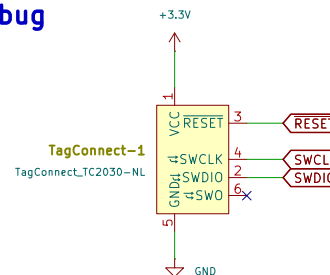
USB C Input



128 Mbit Flash



Debug



The W25Q128JVSQI is a SOIC-8 5x5mm part. The W25Q128JVPQI is a WSON-8 6x5mm part. KiCad has footprints for both

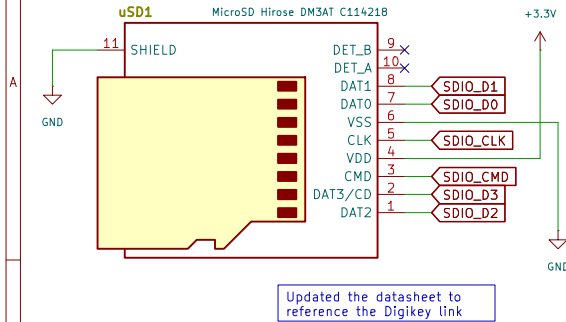
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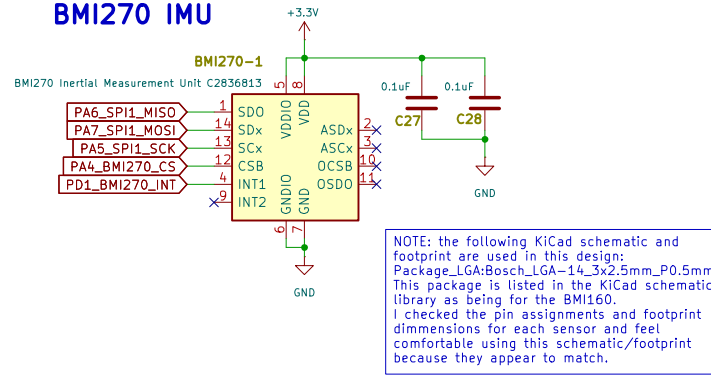
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Id: 1/3

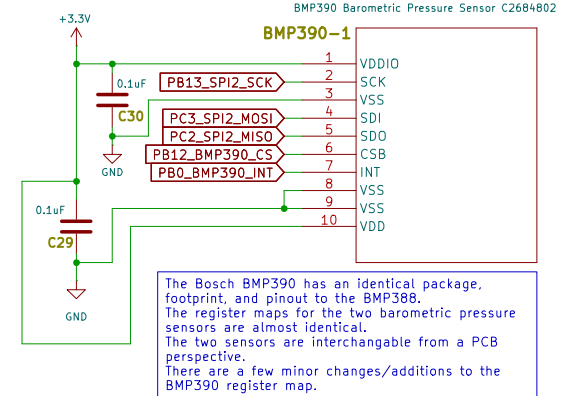
Micro SD Card



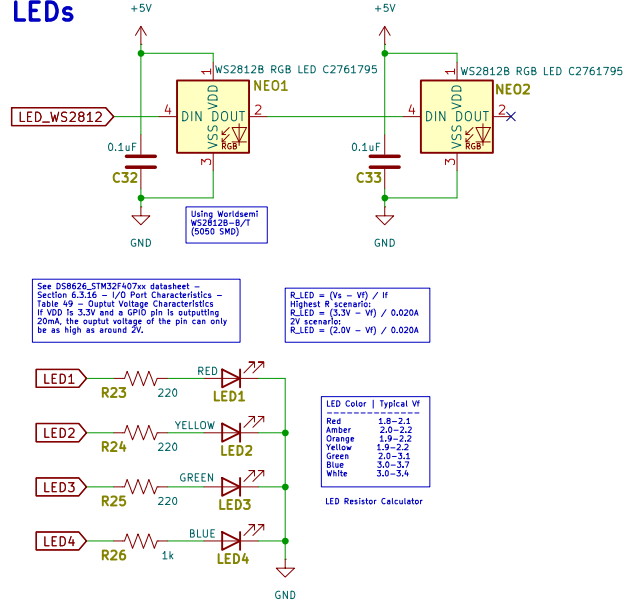
BMI270 IMU



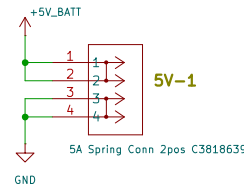
BMP390 Barometric Pressure Sensor



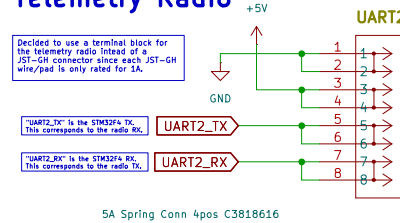
LEDs



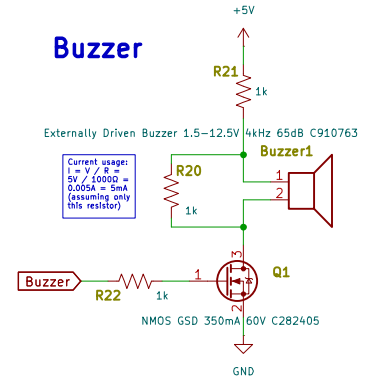
5V External Power



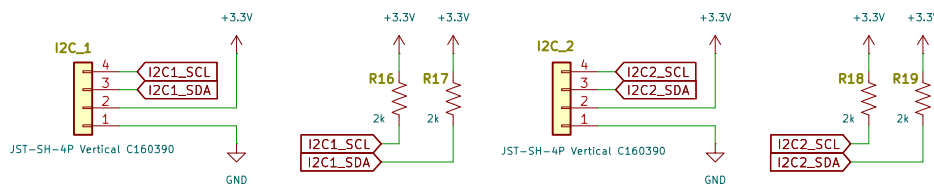
Telemetry Radio



Buzzer



I2C



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KiCad E.D.A. 9.0.2

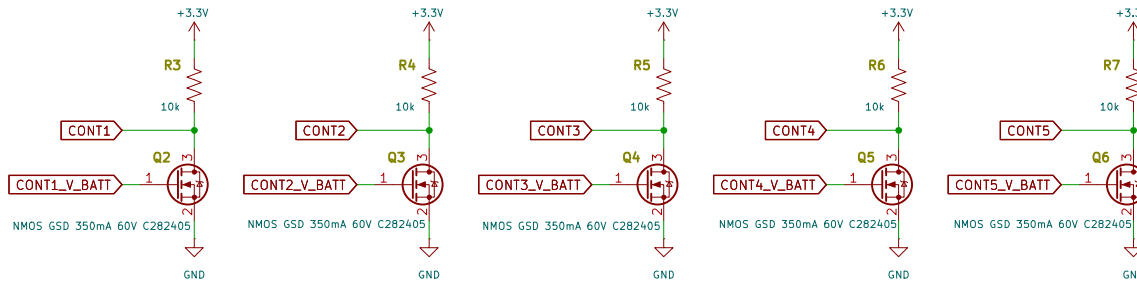
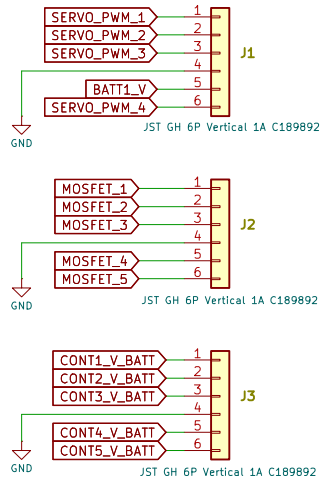
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Servo, Mosfet, and Battery Signals

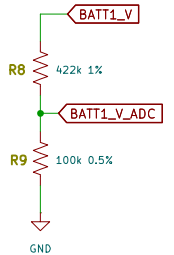
NOTE:
When a CONT# signal is
grounded, there is continuity

NOTE: The Nexperia 2N7002BK-215
Mosfet has a max V_{DS} of 20V.
A voltage divider before the mosfet or a
different mosfet would need to be used
if the battery voltage is more than 20V.

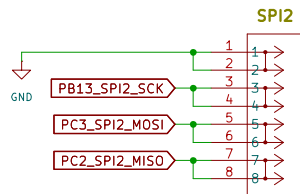


Max battery input voltage: 16.8V
Max ADC pin input voltage: 3.3V (use
3.2V for a small safety factor)
Desired Voltage Divider Ratio: 0.1905
Voltage Divider Ratio chosen:
 $100k / (100k + 422k) = 0.1916$

Each LiPo battery type used to power
the board will have a different number
of discrete voltage steps in the 0 to
3.3V range
4s LiPo (-0.9V ADC voltage range) -
1118 steps
3s LiPo (-0.675V ADC voltage range) -
838 steps
2s LiPo (-0.45V ADC voltage range) -
558 steps



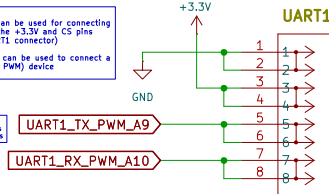
Additional I/O



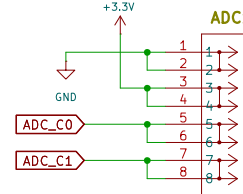
5A Spring Conn 4pos C3818616

NOTE:
the SPI2 connector can be used for connecting
an SPI device (with the +3.3V and CS pins
coming from the UART1 connector)
OR
the UART1 connector can be used to connect a
UART (or low current PWM) device

PAD or PALO
can be used as
chip select pins



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Sheet: /STEVE Page 3/
File: steve_3.kicad_sch

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Id: 3/3